

IBONG TIRIRIT (PIPE 6)

A vertical turbine pump with the jump and motor closed coupled and design to be installed underground, as in the case of a deepwell pump.

- A. Horizontal split case pump
- B. Booster pump
- C. Submersible pump
- D. Vertical shaft turbines pump

Answer: C

An underground formation that contains sufficient saturated permeable material to yield significant quantities of water.

- A. Aquifer
- B. Ground water
- C. Wet pit
- D. Well water

Answer: A

A timber, concrete or masonry enclosure having a screened inlet kept partially filled with water by an open body of water such as pond, lake, or streams.

- A. Aquifer
- B. Ground water
- C. Wet pit
- D. Well water

Answer: C

Water which is available from well, driven into water bearing subsurface strata (aquifer)

- A. Aquifer
- B. Ground water
- C. Wet pit
- D. Well water

Answer: B

Imparts velocity to the liquid, resulting from centrifugal force as the impeller is rotated.

- A. Impeller
- B. Stuffing box
- C. Casing
- D. Shaft sleeve

Answer: A

A means of throttling the leakage which would otherwise occur at the point of entry of the shaft into the casing.

- A. Impeller
- B. Stuffing box
- C. Casing
- D. Shaft sleeve

Answer: B

Protect the shaft where it passes through the stuffing box.

- A. Impeller
- B. Stuffing box
- C. Casing
- D. Shaft sleeve

Answer: D

The ratio of the actual vapor density to the vapor density at saturation

- A. Relative humidity
- B. Absolute humidity
- C. Humidity ratio
- D. Saturation ratio

Answer: A

An expression of the mass of water vapor per unit mass of dry air

- A. Relative humidity
- B. Absolute humidity
- C. Humidity ratio
- D. Saturation ratio

Answer: C

Which of the following is equivalent to 1lb?

- A. 5000 grains
- B. 6000 grains
- C. 7000 grains
- D. 8000 grains

Answer: C

The bypass factor for large commercial units

- A. Around 20%
- B. Around 30%
- C. Around 10%
- D. Around 50%

Answer: C

Sometimes called specific humidity

- A. Relative humidity
- B. Absolute humidity
- C. Humidity ratio
- D. Saturation ratio

Answer: C

For any given barometric pressure, the humidity ratio is a function of the

- A. Critical temperature
- B. Dew point temperature
- C. Dry bulb temperature
- D. Wet bulb temperature

Answer: B

The humidity ratio corresponding to any given dew point temperature varies with the total barometric pressure, increasing as the barometric pressure

- A. Increases
- B. Decreases
- C. Varies
- D. Constant

Answer: B

The ratio of the mass of water vapor in the air per unit mass of dry to the mass of water vapor requires for saturation of the same air sample

- A. Relative humidity
- B. Absolute humidity
- C. Humidity ratio

D. Saturation ratio

Answer: D

A closed channel excavated through an obstruction such as a ridge of higher land between the dam and the powerhouse is called

- A. Canal
- B. Headrace
- C. Penstock
- D. Tunnel

Answer: D

The small reservoir in which the water level rises or falls to reduce the pressure swings so that they are not transmitted to the closed conduit is called

- A. Penstock
- B. Power reservoir
- C. Pressure tank
- D. Tunnel

Answer: D

The regulation of the water in the form of a relatively small pond or reservoir provided at the plant is called

- A. Pondage
- B. Water storage
- C. Reservoir
- D. Lake

Answer: A

A structure used to relieve the reservoir of excess water

- A. Spillway
- B. Diversion channel
- C. Butress dam
- D. Arch dam

Answer: A

The fluid pressure difference created by column of heated gas, as by chimney is called

- A. Natural draft
- B. Forced draft
- C. Induce draft
- D. Required draft

Answer: A

The pressure rise from greater to a lesser vacuum, created in a gas loop between stream generator outlet and chimney by means of a fan is called

- A. Forced draft
- B. Induced draft
- C. Required draft
- D. Balanced draft

Answer: B

Also known as Francis turbines or radial flow turbines

- A. Impulse turbines
- B. Tangential turbines
- C. Reaction turbines
- D. Axial flow turbines

Answer: C

The turbine, draft tube and all related parts comprise what is known as

- A. Powerhouse
- B. Forbay
- C. Setting
- D. Surge chamber

Answer: C

When a forbay is not part of the generating plant's design, it will be desirable to provide a _____ in order to relieve the effects of rapid changes in flowrate.

- A. Forbay
- B. Draft tube
- C. Surge chamber
- D. Penstock

Answer: C

To keep the deflected jet out of the way of the incoming jet, the actual angle is limited to approximately

- A. 90 deg.
- B. 135 deg.
- C. 165 %
- D. 175 deg.

Answer: C

The minimum fluid energy required at the pump inlet for satisfactory operation is known as

- A. NPSHR
- B. NPSHA
- C. Velocity head
- D. Friction head

Answer: A

Throttling the input line to a pump and venting or evacuating the receiving tank

- A. Both increase cavitation
- B. Both decrease cavitation
- C. Both eliminate cavitation
- D. Both drive cavitation

Answer: A

Traditional reciprocating pumps with pistons and rods can be either single-acting or double acting and are suitable up to approximately

- A. 2000 psi
- B. 4000 psi
- C. 8000 psi
- D. 10,000 psi

Answer: A

In a sensible heating process the final humidity ratio is:

- a. Increased
- b. Decreased
- c. The same
- d. Cannot be determined

Answer: C

In an adiabatic saturation process the final relative humidity is:

- A. Increased
- B. Decreased
- C. The same
- D. Cannot be determined

Answer: A

Cooling and dehumidifying process is commonly used for:

- A. Summer air conditioning
- B. Ice making
- C. Winter cooling
- D. Heat pumping

Answer: A

An adiabatic saturation process is also known as:

- A. Constant WB temperature process
- B. Constant DB temperature process
- C. Constant DP temperature process
- D. Constant vapor temperature process

Answer: A

The process of simultaneous heating and dehumidifying is known as:

- A. Sensible heating
- B. Cooling and dehumidifying
- C. Sensible cooling
- D. Chemical dehumidifying

Answer: D

The process in increasing the dry-bulb temperature without changing the humidity ratio is known as:

- A. Sensible heating
- B. Cooling and dehumidifying
- C. Sensible cooling
- D. Heating and dehumidifying

Answer: A

The process of simultaneous cooling and decreasing of humidity ratio is known as:

- A. Sensible cooling
- B. Cooling and humidifying
- C. Cooling and dehumidifying
- D. Heating and dehumidifying

Answer: C

The process of cooling without changing the humidity ratio is known as:

- A. Sensible heating
- B. Cooling and dehumidifying
- C. Sensible cooling
- D. Heating and humidifying

Answer: C

The process of simultaneous heating and humidifying ratio is known as:

- A. Sensible cooling
- B. Cooling and humidifying
- C. Heating and humidifying
- D. Heating and dehumidifying

Answer: C

Which of the following cycle is bulky and involves toxic fluids hence it is unsuitable for home and autocooling

- A. Carnot refrigeration cycle
- B. Absorption cycle
- C. Vapor and compression cycle
- D. Air refrigeration cycle

Answer: B

Which of the following refrigeration cycle is practical when large quantities of waste or inexpensive heat energy are available?

- A. Heat driven refrigeration cycle
- B. Absorption cycle
- C. Vapor compression cycle
- D. Air refrigeration cycle

Answer: A

A product of computer-aided manufacturing and precision machining were introduced commercially in the late 1980's as replacements for reciprocating compressors in small residential air conditioners

- A. Reciprocating compressors
- B. Centrifugal compressors
- C. Rotary compressors
- D. Scroll compressors

Answer: D

Condensers used in small and medium sized up to approximately 100 tons refrigerators.

- A. Air-cooled condensers
- B. Water cooled condensers
- C. High side condensers
- D. Low side condensers

Answer: A

For efficient operation, the condensing temperature should not be lower than

- A. 5 C
- B. 17 C
- C. 10 C
- D. 20 C

Answer: A

For efficient operation, the condensing temperature should not be more than

- A. 5 C
- B. 17 C
- C. 10 C
- D. 20 C

Answer: B

An evaporator in a refrigeration unit makes use of which heat transfer modes?

- A. Conduction
- B. Convection
- C. Radiation
- D. All of the above

Answer: D

Energy added to a vapor is known as a latent heat of vaporization

- A. Latent heat of vaporization
- B. Sensible heat of vapor
- C. Superheat
- D. B and C

Answer: C

The vaporization process that occurs at temperature below the triple point of a substance is called.

- A. Evaporation
- B. Boiling
- C. Sublimation
- D. Condensation

Answer: C

As the pressure increases, the amount of work it can do increases and its enthalpy

- A. Increases
- B. Remained the same
- C. Decreases
- D. Pressure and enthalpy has no relation at all

Answer: A

During the compression process, the internal energy of the refrigerant vapor

- A. Increases
- B. Decreases
- C. Remain the same
- D. Internal energy has no relation to the compression process

Answer: A

The measure of the amount of thermal energy transfer occurring within the evaporator per unit mass under stated conditions.

- A. Refrigeration capacity
- B. Heat capacity
- C. Refrigeration effect
- D. Heat absorptivity

Answer: C

The peak power that can be produced on an occasional basis

- A. Brakepower
- B. Intermittent rating
- C. Continuous duty rating
- D. Power rating

Answer: B

A value of a property that includes the effect of friction is known as:

- A. Brake value
- B. Indicated value
- C. Friction value
- D. Actual value

Answer: A

Forced draft fans are run at relatively high speeds in the range of

- A. 1200 to 1800 rpm
- B. 1500 to 2000 rpm
- C. 1000 to 1600 rpm
- D. 900 to 1500 rpm

Answer: A

Chimneys that rely on natural draft are sometimes referred to as

- A. Natural chimney
- B. Normal chimney
- C. Gravity chimney
- D. Stack

Answer: C

For realistic problems, the achievable stack effect probably should be considered to be

- A. 75% of the ideal
- B. Equal to the ideal
- C. 80 % of the ideal
- D. Half the ideal

Answer: C

In a balanced system, the available draft is

- A. Unity
- B. 100
- C. Zero
- D. Infinite

Answer: C

It is a device used for atomizing or cracking fuel oil and through which the fuel oil is injected into the working cylinders of Diesel engines

- A. Atomizer
- B. Injector
- C. Fuel spray nozzle
- D. Cracker

Answer: A

The device that transfer thermal energy from one fluid to another

- A. Condenser
- B. Feedwater
- C. Evaporator
- D. Heat exchanger

Answer: D

Undergrounds system of hot water and / or steam

- A. Hydrothermal resource
- B. Geothermal resource
- C. Hot water resource
- D. High pressure water system

Answer: A

The process of returning spent geothermal fluids to the subsurface is called

- A. Injection
- B. Returning
- C. Rejection
- D. Exhaustion

Answer: A

The simultaneous demand of all customers required at any specified point in an electric power system is called

- A. Demand
- B. Electrical demand
- C. Power demand
- D. Load

Answer: D

The electricity generating plants that are operated to meet the peak or maximum load on the system are called

- A. Peaking plants
- B. High-load plants
- C. High demanding plants
- D. Add-on plants

Answer: A

The capacity of a substance to transmit a fluid is called

- A. Fluidity
- B. Permeability
- C. Porosity
- D. Smoothness

Answer: B

The ratio of the aggregate volume pore spaces in rock or soil to its total volume is called

- A. Porosity
- B. Sphericity
- C. Permeability
- D. Salinity

Answer: A

Cooling efficiency is typically

- A. 75 to 80%
- B. 50 to 70%
- C. 80 to 85%
- D. 90 to 95%

Answer: B

Forced draft towers can cool the water to within

- A. 10 to 12 F
- B. 5 to 6 F
- C. 20 to 24 F
- D. 2.5 to 3 F

Answer: B

The higher the wet bulb temperatures

- A. The higher the efficiency
- B. The lower the efficiency
- C. The efficiency is limited
- D. The efficiency is maximum

Answer: B

Define the relative difficulty in cooling, essentially the relative amount of contact area or fill volume required

- A. Heat load
- B. Tower load
- C. Cooling duty
- D. Rating factor

Answer: D

Evaporation loss can be calculated from the humidity ratio increase and is approximately _____ decrease in water temperature

- A. 0.18% per C
- B. 0.25% per C
- C. 0.10% per C
- D. 0.30% per C

Answer: A

Water lost in small droplet and carried away by the air flow

- A. Range
- B. Approach
- C. Drift
- D. Bleed-off

Answer: C

The ratio of total dissolved solids in the recirculating water to the total dissolved solids in the make-up water.

- A. Ratio of concentration
- B. Cooling efficiency
- C. Coil efficiency
- D. Bypass factor

Answer: A

A dry-cooling tower where steam travels through large diameter "trunks" to a cross-flow heat exchanger where it is condensed and cooled by the cooler air

- A. Direct condensing tower
- B. Indirect condensing dry cooling tower
- C. Evaporative cooling tower
- D. Atmospheric cooling tower

Answer: A

The minimum fluid energy required at the pump inlet for satisfactory operation is known as

- A. NPSHR
- B. NPSHA
- C. Velocity head
- D. Friction head

Answer: A

The actual fluid energy at the inlet

- A. NPSHR
- B. NPSHA
- C. Velocity head
- D. Friction head

Answer: B

Throttling the input line to a pump and venting or evacuating the receiver tank

- A. Both increase cavitation
- B. Both decrease cavitation
- C. Both eliminate cavitation
- D. Both drive cavitation

Answer: A

Traditional reciprocating pumps with pistons and rods can be either single-acting or double acting and are suitable up to approximately

- A. 2000 psi
- B. 4000 psi
- C. 8000 psi
- D. 10,000 psi

Answer: A

Plunger pumps are only single-acting and are suitable up to approximately

- A. 2000 psi
- B. 4000 psi
- C. 8000 psi
- D. 10,000 psi

Answer: D

The ratio of the actual to the ideal heat transfer coefficient.

- A. Fouling factor
- B. Sensible heat ratio
- C. Cleanliness factor
- D. Biot number

Answer: C

The ratio of the maximum to minimum mass steam flow rates at which the temperature can be accurately contained by the desuperheater.

- A. Cleanliness factor
- B. Capacity factor
- C. Turndown ratio
- D. Fouling factor

Answer: C

With a reversible regenerator, the thermal efficiency of the Ericsson cycle is _____ to that of the Carnot cycle

- A. Less than
- B. Greater than
- C. Proportional
- D. Equal

Answer: D

The electro magnetic radiation emitted from the daughter nucleus when an alpha particles leaves the parent atom

- A. Neutron
- B. Positron
- C. Gamma ray
- D. K-capture

Answer: C

Which of the following statement is a scheme for energy storage?

- A. Pumped hydro
- B. Magnetic energy storage
- C. Thermal energy storage
- D. All of the above

Answer: D

A reactor that employs fast or high energy neutrons and contains no moderator is called

- A. High energy reactor
- B. Fast reactor
- C. High speed reactor
- D. Turbo reactor

Answer: B

A reactor that utilizes slow moving neutrons

- A. Slow reactor
- B. Intermediate reactor
- C. Thermal reactor
- D. Slow speed reactor

Answer: C

The water bearing stratum of permeable sand, rock, or gravel is called

- A. Reservoir
- B. Water source
- C. Aquifer
- D. Well

Answer: C

The electricity generating units that are operated to meet the constant or minimum load on the system is called

- A. Constant load plants
- B. Baseload plants
- C. Invariable load plants
- D. Steady load plants

Answer: B

The effectiveness of a body as a thermal radiator at a given temperature,

- A. Absorptivity
- B. Conductivity
- C. Emissivity
- D. Reflectivity

Answer: C

What are the main components in a combined cycle power plant?

- A. Diesel engine and air compressor
- B. Steam boiler and turbine
- C. Gas engine and waste heat boiler
- D. Nuclear reactor and steam boiler

Answer: C

The size of a steam reciprocating pumps is generally designed by a three-digits number size as 646. The first digit designates

- A. Stroke of the pump in inches
- B. Inside diameter of the steam cylinder measured in inches
- C. Percent clearance
- D. Number of cylinder

Answer: B

The power required to deliver a given quantity of fluid against a given head with no losses in the pump is called

- A. Wheel power
- B. Brake power
- C. Hydraulic power
- D. Indicated power

Answer: C

Fluid that are pumped in processing work are frequently more viscous than water. Which of the following statement is correct?

- A. Reynolds number varies directly as the viscosity
- B. Efficiency of a pump increases as the viscosity increases
- C. Increased fluid friction between the pump parts and the passing fluid increases useful work
- D. Working head increases as the viscosity

Answer: C

The law that states "Entropy of all perfect crystalline solids is zero at absolute zero temperature"

- A. Zeroth law of thermodynamic
- B. First law of thermodynamics
- C. Second law of thermodynamics
- D. Third law of thermodynamics

Answer: D

An expression of the mass of water vapor per unit mass of dry air

- A. Relative humidity
- B. Absolute humidity
- C. Humidity ratio
- D. Saturation ratio

Answer: C

Which of the following is equivalent to 1 lb?

- A. 5000 grains
- B. 6000 grains
- C. 7000 grains
- D. 8000 grains

Answer: C

The locus of states that the same value of stagnation enthalpy and mass flux is called.

- A. Fanno line
- B. Rayleigh line
- C. Willian's line
- D. Mollier's line

Answer: A

Combining the conservation of mass and momentum equations into a single equation and plotting it on the h-s diagram yield a curve called

- A. Fanno line
- B. Rayleigh line
- C. Willian's line
- D. Mollier's line

Answer: B

Across the shock, the stagnation temperature of an ideal gas

- A. Increases
- B. Decreases
- C. Remains constant
- D. Proportional

Answer: C

Which of the following types of air dryers works by absorbing moisture on a solid desiccant or drying material such as activated alumina, silicon gel or molecular sieve?

- A. Regenerative dryer
- B. Spray dryer
- C. Deliquescent dryer
- D. Refrigerated dryer

Answer: C

When two bodies, isolated from other environment, are in thermal equilibrium with a third body, the two are in thermal equilibrium with each other.

- A. Zeroth law of thermodynamic
- B. First law of thermodynamics
- C. Second law of thermodynamics
- D. Third law of thermodynamics

Answer: A

The sum of the energies of all the molecules in a system where energies appear in several complex forms is the

- A. Kinetic energy
- B. Internal energy
- C. Potential energy
- D. Friction energy

Answer: B

The Carnot refrigeration cycle includes all of the following process except

- A. Isentropic expansion
- B. Isenthalpic expansion
- C. Isothermal expansion
- D. Isentropic compression

Answer: B

The maximum possible work that can be obtained a cycle operating between two reservoirs is found from

- A. Process irreversibility
- B. Carnot efficiency
- C. Availability
- D. Reversible work

Answer: B

The following factors are necessary to define a thermodynamic cycle expect

- A. The working substance
- B. High and low temperature reservoirs
- C. The time it takes to complete the cycle
- D. The means of doing work on the system

Answer: C

A temperature above which a given gas cannot be liquefied

- A. Cryogenic temperature
- B. Absolute temperature
- C. Vaporization temperature
- D. Critical temperature

Answer: D

The geometrical electricity generating plants that employs a closed-loop heat exchange system in which the heat of the primary fluid is transferred to a secondary fluid, which is thereby vaporized and used to drive a turbine generator set

- A. Binary cycle plant
- B. Dual cycle plant
- C. Double cycle plant
- D. Cascade cycle

Answer: A

The geothermal solution which contains appreciable amounts of sodium chloride or other salts is called

- A. Sulfur dioxide
- B. Potassium silicate
- C. Sea water
- D. Brine

Answer: D

The quantity of heat required to raise the temperature of one pound of water one degree Fahrenheit at standard conditions.

- A. Btu
- B. Calorie
- C. Joule
- D. Centigrade

Answer: A

The process that uses a stream of geothermal of hot water or steam to perform successive task requiring temp is called

- A. Diminishing heat
- B. Decreasing heat
- C. Cascading heat
- D. Negative gradient process

Answer: C

The water formed by condensation of steam is called

- A. Distilled water
- B. Condensate
- C. Dew
- D. Condenser

Answer: B

A structure that removes heat from condensate is called

- A. Desuperheater
- B. Cooling tower
- C. Evaporator
- D. Condenser

Answer: B

In a typical hydroelectric generating plant using reaction turbines, the turbine is generally housed in a

- A. Powerhouse
- B. Penstock
- C. Forbay
- D. Setting

Answer: A

After the water passes through the turbine, it is discharged through the draft tube to the receiving reservoir known as the

- A. Tail race
- B. Tailwater
- C. Draft tube
- D. Setting

Answer: B

If a draft tube is not employed, water maybe returned to the tailwater by way of a channel is known as the

- A. tail race
- B. tailwater
- C. draft tube
- D. setting

Answer: A

The turbine, draft tube and all related parts comprise what is known as

- A. Powerhouse
- B. Forbay
- C. Setting
- D. Surge chamber

Answer: C

When a forbay is not part of the generating plant's design, it will be desirable to provide a _____ in order to relieve the effects of rapid changes in flowrate.

- A. Forbay
- B. Draft tube
- C. Surge chamber
- D. Penstock

Answer: C

To keep the deflected jet out of the way of the incoming jet, the actual angle is limited to approximately

- A. 90 deg.
- B. 135deg.
- C. 165%
- D. 175deg.

Answer: C

The spontaneous vaporization of the fluid, resulting in a degradation of pump performance.

- A. Cavitation
- B. Vapor lock
- C. Available head
- D. Net head

Answer: A

A dry cooling tower where steam is condensed by cold water jets (surface or jet condenser)

- A. Direct condensing tower
- B. Indirect condensing dry cooling tower
- C. Evaporative cooling tower
- D. Atmospheric cooling tower

Answer: B

If the heat is being removed from water, the device is known as

- A. Chiller
- B. Cooler
- C. Air conditioner
- D. Air cooler

Answer: A

Which of the following is the refrigerant "of choice" in uniting air conditioners?

- A. R-22
- B. R-123
- C. R-11
- D. R-502

Answer: A

In new equipment, which of the following replace R-11?

- A. R-12
- B. R-123
- C. R-502
- D. R-22

Answer: B

The mass flow rate produced by the compressor is

- A. Proportional to the mass of the suction vapor that the compressor takes in at the suction inlet per unit time
- B. Equal to the mass of the suction vapor that the compressor takes in at the suction inlet per unit time
- C. Less than the mass of the suction vapor that the compressor takes in at the suction inlet per unit time
- D. Greater than the mass of the suction vapor that the compressor takes in at the suction inlet per unit time

Answer: B

A boiler operated at pressure not exceeding 1.055 kg/cm² gage steam, or water temperature not exceeding 121°C.

- A. Low pressure heating boiler
- B. Hot water supply boiler
- C. Miniature
- D. Power boiler

Answer: A

The very hot steam that doesn't occur with a liquid

- A. Pure steam
- B. Saturated vapor
- C. Critical steam
- D. Dry steam

Answer: D

The steam produced when the pressure on a geothermal liquid is reduced

- A. Low pressure steam
- B. High quality steam
- C. Sub pressure steam
- D. Flashed steam

Answer: D

The vent or hole in the Earth surface, usually in a volcanic region, from which, gaseous vapors, or hot gases issue.

- A. Fumaroles
- B. Crater
- C. Hot spot
- D. Hot spring

Answer: A

The earth interior heat made available to man by extracting it from hot water or rocks.

- A. Geological heat
- B. Geothermal heat
- C. Volcanic heat
- D. None of these

Answer: B

Device that takes advantage of the relatively constant temperature of the Earth's interior, using it as a source and sink of heat for both heating and cooling.

- A. Geothermal devices
- B. Geothermal generator
- C. Geothermal heat pumps
- D. Geothermal turbines

Answer: C

The spring that shoots jets of hot water and steam into the air is called

- A. Geyser
- B. Hot jet
- C. Thermal jet
- D. Geyser

Answer: A

What is the power seat width of a spray valve?

- A. 1/16 in.
- B. 1/32 in.
- C. 1/8 in.
- D. 1/4 in.

Answer: A

A device which automatically governs or controls the speed of an engine.

- A. Servomotor
- B. Indicator
- C. Governors
- D. Speedometer

Answer: C

The useful energy transfer in Btu/hr divided by input power in watts. This is just the coefficient of performance expressed in mixed units.

- A. Energy efficient ratio
- B. Coil efficient
- C. Bypass factor
- D. Sensible heat ratio

Answer: A

If EER is the energy efficiency ratio, and COP is the coefficient of performance then

- A. $EER = 3.41COP$
- B. $COP = 3.41EER$
- C. $EER \times COP = 3.41$
- D. $3.41EER \times COP = 1$

Answer: A

Which of the following is a reversed Rankine vapor cycle?

- A. Carnot refrigeration cycle
- B. Vapor refrigeration cycle
- C. Air refrigeration cycle
- D. Absorption cycle

Answer: B

The most common type of refrigeration cycle, finding application in household refrigerators, air conditioners for cars and houses, chillers and so on.

- A. Carnot refrigeration cycle
- B. Vapor refrigeration cycle
- C. Air refrigeration cycle
- D. Absorption cycle

Answer: B

Which of the following is a reversed Brayton cycle?

- A. Carnot refrigeration cycle
- B. Vapor refrigeration cycle
- C. Air refrigeration cycle
- D. Absorption cycle

Answer: C

A 100kg is at 0°C is heated by supplying 2000KJ of heat to it. If the heat of fusion is 335kJ/kg how many kilograms of ice will melt into water?

- A. 8 kg
- B. 10 kg
- C. 6 kg
- D. 3 kg

Answer: C

Determine the quantity of latent heat transferred to an evaporator when 5 kg of water undergoes those changes from a liquid at 0°C to ice at 0°C .

- A. 1,300 kJ
- B. 1,675 kJ
- C. 1,450 kJ
- D. 1,800 kJ

Answer: B

Compute the rate of water transfer by melting an ice at a rate of 50 kg/hr.

- A. 8 kJ/sec
- B. 10 kJ/sec
- C. 5 kJ/sec
- D. 12 kJ/sec

Answer: C

10 kg of water at 60°C receives 42,000kJ of heat coming from the surrounding. Assuming it is open to the atmosphere, how many kilograms of water will be vaporized?

- A. 18 kg
- B. 15 kg
- C. 10 kg
- D. 20 kg

Answer: A

As the liquid changes phase to vapor, its enthalpy

- A. Increases
- B. Decreases
- C. Remains constant
- D. Enthalpy has no relation to phase change

Answer: A

Which of the following processes does not alter the kinetic energy level of a substance?

- A. Fusion
- B. Adding sensible heat to a solid
- C. Adding sensible heat to a liquid
- D. Superheating a vapor

Answer: A

Condensers used for larger capacities refrigerators.

- A. Air-cooled condensers
- B. Water-cooled condensers
- C. High side condensers
- D. Low side condenser

Answers: B

Coolers and chillers for water generally operate with an average temperature difference of

- A. 3 to 11 $^{\circ}\text{C}$
- B. 5 to 8 $^{\circ}\text{C}$
- C. 6 to 22 $^{\circ}\text{C}$
- D. 10 to 16 $^{\circ}\text{C}$

Answer: A

To avoid freezing problems, entering refrigerant should be

- A. Below -2°C
- B. Above -2°C
- C. Equal to -2°C
- D. 0°C

Answer: B

Suction lines should not be sized too large, as reasonable velocity is needed to carry oil from the evaporator back to the compressor. For horizontal suction lines, the recommended minimum velocity is

- A. 3.8 m/s
- B. 6.1 m/s
- C. 7.1 m/s
- D. 4.8 m/s

Answer: A

Type of turbine used for low heads, high rotational speeds and larger flow rates

- A. Axial flow turbines
- B. Reaction turbines
- C. Radial flow turbines
- D. Impulse turbines

Answer: A

To keep the deflected jet out of the way of the incoming jet, the actual angle is limited to approximately

- A. 135 deg.
- B. 150 deg.
- C. 165 deg.
- D. 175 deg.

Answer: C

Which of the following turbines are centrifugal pumps operating in reverse?

- A. Reaction turbines
- B. Impulse turbines
- C. Tangential turbines
- D. Axial flow turbines

Answer: A

Weight per unit volume is termed as:

- A. Specific gravity
- B. Specific density
- C. Weight density
- D. Pressure

Answer: C

Measure of hotness or coldness of the body

- A. Pressure
- B. Temperature
- C. Energy
- D. Entropy

Answer: B

The temperature at which the water vapor in the air begins to condensed, or the temperature at which the relative humidity of air becomes 100%.

- A. Flash point
- B. Boiling point
- C. Dew point
- D. Freezing point

Answer: C

It consists of a wet and dry bulb thermometers mounted on a strip of metal

- A. Manometer
- B. Gyrometer
- C. Pyrometer
- D. Sling psychrometer

Answer: D

A manometer is an instrument that is used to measure:

- A. Air pressure
- B. Condensate water level
- C. Heat radiation
- D. Air volume

Answer: A

The relationship of water vapor in the air at the dew point temperature to the amount that should be in the air if the air were saturated at the dry-bulb temperature is:

- A. Partial pressure actual dew point
- B. Percentage humidity
- C. Relative humidity
- D. Run faster

Answer: A

All of the following statements about wet bulb temperature are true, EXCEPT

- A. Wet bulb temperature equals adiabatic saturation temperature
- B. Wet bulb temperature is the only temperature necessary to determine grains water per lb of dry air
- C. Wet bulb temperature lies numerically between dew point and dry bulb temperature for unsaturated systems
- D. Wet bulb temperature equals both dry-bulb and dew point temperature at 100% relative humidity

Answer: C

Heat added to or removed from a substance that cause a change of temperature:

- A. Absolute heat
- B. Latent heat
- C. Specific heat
- D. Sensible heat

Answer: D

Plunger pumps are only single-acting and are suitable up to approximately

- A. 2000 psi
- B. 4000 psi
- C. 8,000 psi
- D. 10,000 psi

Answer: D

The ratio of the actual to the ideal heat transfer coefficient.

- A. Fouling factor
- B. Sensible heat ratio
- C. Cleanliness factor
- D. Biot number

Answer: C

The ratio of the maximum to minimum mass steam flow rates at which the temperature can be accurately contained by the desuperheater.

- A. Cleanliness factor
- B. Capacity factor
- C. Turndown ratio
- D. Fouling factor

Answer: C

Used to describe the act of blowing the exhaust product out with the air-fuel mixture

- A. Supercharging
- B. Scavenging
- C. Honing
- D. Choking

Answer: B

Which of the following compresses and increases the amount of air that enters the cylinder per stroke

- A. Supercharging
- B. Scavenging
- C. Honing
- D. Choking

Answer: A

A form of supercharging in which the exhaust gases drive the supercharger

- A. Lugging
- B. Honing
- C. Turbocharging
- D. Blower charging

Answer: C

The maximum speed of the turbine under no and no governing action is called

- A. Runaway speed
- B. Pre governing speed
- C. Governing speed
- D. No load governing

Answer: A

Which of the following statements is false when gasoline is used as diesel fuel.

- A. Gasoline does not ignite
- B. Gasoline wears the fuel-injection pumps because of its low viscosity
- C. Addition of lubrication oil to gasoline will just severe the situation
- D. All of the above

Answer: C

Which of the following is an advantage of hydro-plants?

- A. Hydro plants provide ancillary benefits like irrigation, flood control, aquaculture navigation etc.
- B. Modern hydro generators give high efficiency over considerable range of load. This helps in improving the system
- C. Due to its great ease of taking up and throwing off the load, the hydro power can be used as the ideal spinning reserve in a system mix of the thermal, hydro and power station
- D. All of the above

Answer: B

The hole area behind the dam draining into a stream or river across which dam has been constructed is called

- A. Lake
- B. Forebay
- C. Catchment area
- D. Reserve potential

Answer: C

Dams constructed primarily to store flood waters called

- A. Storage dams
- B. Diversion dams
- C. Detention dam
- D. None of the above

Answer: C

A channel which lead water to a turbine is called

- A. Canal
- B. Tailrace
- C. Penstock
- D. Headrace

Answer: D

Sometimes called percentage humidity

- A. Relative humidity
- B. Absolute humidity
- C. Humidity ratio
- D. Saturation ratio

Answer: D

When measuring _____ of air, the bulb of the thermometer should be shaded to reduce the effect of direct radiation

- A. Dry bulb temperature
- B. Wet bulb temperature
- C. Critical temperature
- D. Saturation temperature

Answer: A

An ordinary thermometer whose bulb is enclosed in a wetted cloth sac or wick

- A. Dry bulb thermometer
- B. Wet bulb thermometer
- C. Ordinary thermometer
- D. Mercury thermometer

Answer: B

To obtain an accurate reading with a wet bulb thermometer, the wick should be saturated with

- A. Clean air
- B. Vapor
- C. Moist
- D. Clean water

Answer: D

The amount by which the wet bulb temperature is reduced below the dry bulb temperature depends on the relative humidity of the air and is collective

- A. Range
- B. Approach
- C. Drify
- D. Wet bulb depression

Answer: D

The measure of the relationship between the dry bulb and dewpoint temperature of the air and as such, it provides a convenient means of Dewpoint temperature of the air when the dry bulb temperature is known

- A. Wet bulb temperature
- B. Saturation temperature
- C. Humidity ratio
- D. Relative humidity

Answer: A

An index of the total heat (enthalpy) of the air

- A. Wet bulb temperature
- B. Saturation temperature
- C. Dry bulb temperature
- D. Dewpoint temperature

Answer: A

The sensible heat of the air is a function of the

- A. Wet bulb temperature
- B. Dry bulb temperature
- C. Dewpoint temperature
- D. Saturation temperature

Answer: B

The latent heat of the air is a function of the

- A. Wet bulb temperature
- B. Dry bulb temperature
- C. Dewpoint temperature
- D. Saturation temperature

Answer: C

The total heat of air is a function of the

- A. wet bulb temperature
- B. dry bulb temperature
- C. dewpoint temperature
- D. saturation temperature

Answer: A

A single suction pump having its suction nozzle on the opposite side casing from the stuffing box and having the face of the suction perpendicular to the longitudinal axis of the shaft.

- A. Centifugal pump
- B. In line pump
- C. End suction pump
- D. Horizontal pump

Answer: C

A centrifugal pump with one or more impellers discharging into one or more bowls and a vertical educator or column pipe used to connect the bowls to the discharge head on which the pump driver is mounted.

- A. Horizontal split case pump
- B. In line pump
- C. End suction pump
- D. Vertical shaft turbine pump

Answer: D

A pump that takes suction from a public service main or private use water system for the purpose of increasing the effective water pressure.

- A. Horizontal split case pump
- B. Booster pump
- C. Submersible pump
- D. Vertical shaft turbine pump

Answer B